

OTT Viewer Engagement EDA

Hotstar Business Scenario

ANALYSIS REPORT	PORTFOLIO PROJECT
33,171 episode records	23 analytical variables
Python EDA	Business-focused insights

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Exploratory data analysis of a synthetic OTT viewer-behaviour dataset. Hotstar is used only as a business scenario; the dataset is not proprietary Hotstar data.

1. Executive Summary

This project investigates how episode characteristics and viewer-behaviour signals relate to engagement, drop-off probability and retention risk. The analysis is designed as a decision-support exercise for an OTT content team, using descriptive EDA and correlation analysis rather than machine learning.

Metric	Verified result	Interpretation
Dataset size	33,171 x 23	Large episode-level analytical sample
Missing / duplicates	0 / 0	No missing-value or exact-duplicate remediation required
Average watch %	56.81%	Typical episode is watched a little over halfway
Average duration	37.66 min	Useful baseline for content-length comparisons
Top engagement genre	Comedy - 77.38	Comedy leads on the project-defined engagement index
Highest drop-off genre	Drama - 24.19%	Drama deserves closer content-level investigation
High retention risk	14.48%	Meaningful subset of episodes carries elevated risk

The strongest relationship in the verified findings is between watch percentage and drop-off probability ($r \approx -0.957$). Cognitive load is also strongly positively associated with drop-off probability ($r \approx 0.793$). These are associations, not proof of causation.

2. Business Problem & Objectives

The business scenario assumes an OTT content team wants to understand why viewers remain engaged with some episodes but lose interest in others. The practical goal is to surface interpretable patterns that can guide content review, pacing decisions and prioritization of high-risk episodes.

- Measure overall viewer engagement and identify high- and low-performing content.
- Examine drop-off probability and retention-risk patterns.
- Compare behaviour across genres, seasons and episodes.
- Assess relationships among hook strength, pacing, cognitive load, watch percentage and drop-off.
- Translate analytical findings into business actions without making unsupported causal claims.

3. Dataset & Methodology

The dataset contains 33,171 episode-level observations and 23 columns. It is synthetic and was created for realistic OTT analysis; therefore, the report does not claim to describe actual Hotstar users or proprietary Hotstar performance.

Stage	Purpose
Validation & cleaning	Inspect schema, types, missing values, duplicates, labels, ranges and safe corrections.
Univariate EDA	Understand individual distributions and baseline behaviour.
Bivariate EDA	Compare viewer and content variables across relevant groups.
Engagement analysis	Use watch behaviour and a project-defined engagement index to rank content.
Drop-off & retention	Identify higher-risk content patterns and segments.
Season / episode analysis	Look for within-title and sequential content patterns.
Correlation analysis	Quantify linear associations among key numerical variables.

Stage	Purpose
Business insights	Convert outputs into decision-oriented findings and recommendations.

Tools used

Python, NumPy, Pandas, Matplotlib and Seaborn. No machine learning, Scikit-learn, Plotly, Power BI, Tableau or additional statistical libraries are used.

4. Key Analytical Findings

Engagement

Comedy records the highest average engagement score at approximately 77.38. The engagement score is a project-defined composite index and should be interpreted as a ranking aid rather than a universal 0-100 metric.

Drop-off

Drama has the highest observed drop-off rate at approximately 24.19%, making it a logical genre for deeper episode-level review rather than an automatic conclusion that drama itself causes abandonment.

Retention

Approximately 14.48% of observations fall into the high retention-risk category, providing a focused pool for content diagnostics.

Hook strength

Hook strength and watch percentage show a moderate positive correlation ($r \approx 0.552$), consistent with stronger openings being associated with more viewing.

Pacing

Pacing score and the composite engagement score show a moderate positive correlation ($r \approx 0.566$). Because pacing is an input to the engagement formula, part of this relationship is mechanical.

Cognitive load

Cognitive load and drop-off probability have a strong positive correlation ($r \approx 0.793$), suggesting cognitively demanding episodes deserve closer review for abandonment risk.

Watch vs drop-off

Watch percentage and drop-off probability show a very strong negative relationship ($r \approx -0.957$), indicating that higher completion behaviour aligns closely with lower modeled drop-off probability.

Episode duration

Episode duration and watch percentage are essentially uncorrelated in the verified result ($r \approx -0.0004$), so duration alone is not a useful explanation for watch percentage in this dataset.

5. Correlation Snapshot

Relationship	r	Reading
Hook strength vs watch %	+0.552	Moderate positive
Pacing vs engagement	+0.566	Moderate positive; partly mechanical
Cognitive load vs drop-off probability	+0.793	Strong positive
Watch % vs drop-off probability	-0.957	Very strong negative
Episode duration vs watch %	-0.0004	Essentially no linear relationship

Correlation is used here as an exploratory measure of association. It does not establish that changing one variable will cause another variable to change.

6. Business Recommendations

Prioritize high-risk episodes

Use the high retention-risk subset as a review queue. Compare pacing, hook strength, cognitive load and watch behaviour before investing in broad catalogue-wide changes.

Investigate Drama at episode level

Drama has the highest observed drop-off rate, but genre-level averages can hide variation. Review titles and episodes with sufficient sample support before acting.

Strengthen early engagement

The positive hook-strength/watch relationship supports testing stronger openings, clearer early stakes and faster delivery of viewer value.

Review cognitively heavy content

Because cognitive load is strongly associated with drop-off probability, examine whether high-load episodes need clearer exposition, pacing adjustments or better narrative signposting.

Do not optimize duration in isolation

The near-zero duration/watch relationship suggests episode length alone is unlikely to be a useful optimization lever in this synthetic dataset.

Use rankings with context

Add sample-size context to top/bottom tables so small groups do not receive disproportionate attention from noisy averages.

7. Limitations & Analytical Guardrails

- The dataset is synthetic; findings are portfolio/educational insights and should not be represented as actual Hotstar customer behaviour.
- Hotstar is the business scenario only and is not the source of the data.
- This is exploratory analysis. Correlations identify associations, not causal effects.
- The engagement score is project-defined. It combines average watch percentage, hook strength, pacing score and visual intensity; its weighting should be treated as an analytical assumption.
- Correlations involving variables that are components of the engagement score can be partly mechanical.
- Group rankings should be interpreted alongside sample sizes and distribution spread, not only means.
- No machine-learning model is used, consistent with the defined project scope.

8. Portfolio & Repository Recommendations

This report fills the communication gap between the code-heavy notebooks and the README. For the strongest portfolio presentation, the repository should expose a clear entry path: README for the quick overview, one consolidated analysis notebook for technical review, and this PDF for stakeholder-facing communication.

Priority	Repository improvement
1	Add/restore the consolidated Hotstar_Viewer_Engagement_EDA.ipynb referenced by the README.
2	Place section notebooks under notebooks/sections/ to reduce root-level clutter.
3	Store this PDF under reports/ and selected charts under reports/figures/.
4	Move raw data to data/raw/ and document the dataset schema/data dictionary.
5	Pin dependency versions and add a LICENSE for stronger reproducibility and reuse clarity.

9. Conclusion

The analysis demonstrates a complete EDA workflow: data validation, behavioural exploration, engagement and drop-off analysis, retention assessment, correlation analysis and business interpretation. The strongest verified signals point toward watch behaviour, cognitive load, hook strength and pacing as useful areas for content-team investigation. The results should be treated as hypotheses from synthetic data, not causal claims or evidence about real Hotstar users.

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